



## François A. Gougeon

### Curriculum vitae

François Gougeon has been involved in digital remote sensing since 1976, when, shortly after receiving his electrical engineering degree with specialisation in digital electronics and microprocessors (B.A.Sc., University of Ottawa, 1976), he became interested in systems of parallel processors for image processing (M.A.Sc., University of Ottawa, 1980).

As a scientist with the Petawawa National Forestry Institute (PNFI) since 1979, his responsibilities have fluctuated between computer system analysis and scientific research, both in digital remote sensing. His first responsibilities were related to the programming, maintenance, and management of the ARIES prototype, an unique system of image analysis, and to assist the digital remote sensing researchers using it. He later became involved in non-supervised classification of satellite images, spectral variations due to spruce budworm damage, and change detection on a series of multi-season Landsat images using an expert system.

In 1984, he was granted a CFS paid educational leave and got a Ph.D. in "Computer Vision". For his Ph.D. (System Engineering, University of Waterloo, 1987), he analysed forest stand texture in medium resolution (1-6m/pixel) aerial images, created an image segmentation process, and examined the effects of view and illumination angles on textural signatures.

Back at PNFI, he started his work on deliniating and recognizing individual tree crowns (ITC) in high spatial resolution (30-100 cm/pixel) aerial images, while also working on an autonomous forestry robot (JACOB). His robotic research followed two streams. The control stream implied the design of electronic circuits and software to coordinate the movements of an autonomous silvicultural robot from its various sensor inputs (e.g., RGB, infrared, sonar). The vision stream implied the analysis and understanding of images taken from inside a forested area ongoing a difficult regeneration (i.e., populated by young conifers dominated by competing vegetation).

In July 1996, he was transferred to the Pacific Forestry Centre (PFC), Victoria, BC. There, his research has been mostly aimed at the development of new methods and processes towards the delineation, identification and regrouping of individual tree crowns (ITCs) from high spatial resolution (30-100 cm/pixel) multispectral images (aerial and satellite). The approaches are based on image processing, statistical analysis and artificial intelligence techniques. Spectral, textural, structural and contextual aspects were considered.

Over time, Dr. Gougeon has developed the **ITC Suite** consisting of about 35 computer programs that embodies this methodology. The system is fairly automatic, requiring people only to provide sample areas with which to "train" the computer in species recognition and then, to assess the results. Also, since aerial LiDAR data or stereo image autocorrelation can be used to produce precise Digital Canopy Model(DCM), this makes it possible to assess forest stand heights and/or ITC heights. Wood volumes can then be calculated the conventional way (via stratification), but with the benefit of more precise species compositions, OR on an ITC basis (or stand basis), as functions of species, crown area, and height.

An important long term goal has been the semi-automatic production of precise forest inventories.

### Publications

Press [here](#), for a publication list.

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